

DATASHEET

Soft-Starter



Main Features

Reference	: SSW900E0604T5E3
Product code	: 14418678
Product family	: SSW900

Basic data

Power	
- Rated current	: 604 A
- Power supply	: 220-575V
- Power supply type	: TT / TN / IT
- Frequency	: 50/60Hz
- Number of controlled phases	: 3 phases
- Integrated bypass	: Yes

Motor power - standard connection (cv/kW)

- 220/230V	: 250 / 185
- 380/400V	: 450 / 330
- 440/460V	: 500 / 370
- 575V	: 650 / 485
- 690V	: No

Motor power - delta inside connection (cv/kW)

- 220/230V	: 450 / 330
- 380/400V	: 750 / 550
- 440/460V	: 850 / 630
- 575V	: 1150 / 860

Electronic and fan	
Supply voltage (control)	: 110-130V
Supply voltage (cooler)	: 110-130V
Connection to the motor	
- Standard (3 leads)	: Yes
- Inside delta (6 leads)	: Yes
Duty cycle	
- Estándar [1]	: AC-53b 3-30:690
- With ventilation kit [2]	: No

Digital inputs

- Quantity (standard)	: 6
- Type	: DC NPN / PNP
- Voltage	: 24 Vdc
- Minimum high level	: 18 Vdc
- Maximum low level	: 3 Vdc
- Maximum voltage	: 30 Vdc
- Input current	: 11 mA @ 24 Vdc
- Function	: Programmable
- Reading via communication	: Yes

Input for motor thermistor

- Quantity (standard)	: 1
- Activation	: 3k9 Ohms
- Deactivation	: 1k6 Ohms
- Minimum resistance	: 100 Ohms

Analog outputs

- Quantity (standard)	: 1
- Level	: 0 to 10 Vdc and 0 to 20 mA
- Resolution	: 10 bits
- Function	: Programmable
- Write via communication	: Yes

Digital outputs

- Quantity	: 3
- Type	: 2 x NO relay, 1 NO/NC relay
- Maximum voltage	: 250 Vac
- Maximum current	: 1 A
- Function	: Programmable
- Write via communication	: Yes

Functions and Features

Control type start and stop	
- Voltage ramp	: Yes
- Voltage ramp and current limit	: Yes
- Current limit	: Yes

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- Current limitation ramp : Yes
- Pump control : Yes
- Torque control : Yes
- Direct on line start : Yes

Functions/Features

Specials functions

- Torque pulse at start (kick start) : Yes
- Jog : Yes
- DC braking (with contactor) : Yes
- Optimal Braking (without contactor) : Yes
- Reverse braking : Yes
- Changing of the rotation direction : Yes
- Multimotor application : No
- Emergency start (Fire Mode) : Yes

Diagnosis

- Latest faults : Yes, LOG
- Starting and cycle diagnosis : Yes
- Last previous faults : Yes
- Last previous alarms : Yes
- Last previous events : Yes

Others

- Set-up of the programmable fan : Yes
- Copy of the parameters via keypad : Yes
- User 1 and user 2 : Yes
- Oriented Start-up : Yes
- Protections actuation in errors and alarms : Yes
- Fault auto reset : Yes
- Real time clock (RTC) : Yes
- SoftPLC (ladder programming) : Yes, 4K bytes

Slots (para accessories options)

- Quantity : 2
- Digital inputs and outputs expansion : No
- PTC inputs : No
- Pt-100 inputs : No
- Current sensor (by extern bypass) : No

Communications (see Communication)

Protections

Voltage

- Power supply phase loss : Yes
- Overvoltage and undervoltage : Yes
- Voltage imbalance : Yes
- Electronic undervoltage (control) : Yes

Current

- Overcurrent and undercurrent : Yes
- Current imbalance : No

Torque

- Overtorque e undertorque : Yes

Potencia

- Overpower and underpower : Yes

Temperature

- Thyristor overheating : Yes
- PTC thermistor motor overheating : Yes
- PT100 thermocouple motor overheating : No

Overload

- Motor overload (thermal class) : Yes
- Thyristor overload : Yes
- Inverted phase sequence : Yes
- External fault : Yes
- Not connected motor : Yes
- Wrong motor connection : Yes
- Frequency out of range : Yes

Ground fault

- Ground fault : No

Bypass

- Bypass not opened : Yes
- Bypass not closed : Yes
- Overcurrent before closing the bypass : Yes
- Undercurrent before closing the bypass : Yes

Time

- Time before starting : Yes
- Time after starting : Yes
- Time between starting : Yes
- Locked rotor at start : Yes

Communications

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- Programming error	: Yes
- Communication faults	: Yes
Potencia	
- Power short circuit with motor switched off	: Yes
- Thyristor fault	: Yes
- Hardware self-diagnosis	: Programmable via optional
Parameterization	
- Parameterization via trimpots and dip switches	: No
- Parameterization via keypad	: Yes
- Parameterization via software	: WPS
Local HMI	
- Type	: Yes, removable
Remote HMI	
- Type	: Local removable
- Frame for HMI	: Optional
- HMI degree of protection	: IP56
- Maximum cable length	: 20 m / 65.6 ft
Local or remote HMI data	
- Display	: LCD graphic
- Keys	: 12 keys
- Password	: Yes
- Language	: Yes
- Help	: Yes
- Copy function	: Yes
- Status graphic (read)	: Yes
- HMI USB	: Yes
Communication	
- Modbus-RTU RS-232	: No
- Modbus-RTU RS-485	: Available with optional
- Modbus-RTU USB	: Yes
- WEG Bus	: No
- Modbus-TCP	: Available with optional
- Profibus-DP	: Available with optional
- Profibus-DPV1	: Available with optional
- Profinet	: Available with optional
- DeviceNet	: Available with optional
- EtherNet/IP	: No
- EtherNet/IP 2P	: Available with optional
- Profinet IO 2P	: Available with optional
- CANopen	: Available with optional
Ambient conditions	
- Enclosure	: IP00
- Pollution degree (UL 508)	: 2
Temperature	
- Minimum	: -10 °C / 14 °F
- Rated	: 40 °C / 104 °F
- Current reduction factor above nominal [3]	: 2% for each degree
Relative humidity (non-condensing)	
- Minimum	: 5 %
- Maximum	: 90 %
Altitude	
- Under rated operating conditions	: 1000 m (3281 ft)
- Maximum allowed with reduction factor [4]	: 4000 m (13123 ft)
- Current reduction factor above 1000 m [5]	: 1 % for each 100 m (328 ft)
- Voltage reduction factor above 2000 m [6]	: 1.1 % for each 100 m (328 ft)
Dimensions	
- Frame	: E
- Height	: 575 mm / 22.6 in
- Width	: 390 mm / 15.4 in
- Depth	: 260 mm / 10.2 in
- Weight	: 38 kg / 83.8 lb
Sustainability Directives	
- RoHS	: Yes
- Conformal coating	: 3C2
Standards Compliance	
- Safety	: UL 508
- Low voltage	: EN 60947-4-2
- Overvoltage category	: III (UL508 / EN61010)
- EMC	: Directive 89/336/EEC
Certifications	
- Product certification	: UL, CE

Notes

1) Starting duty cycle AC-53b 3-30:690:



- 3 x Rated current of the SSW900 during ;
 - ;
 - 100% of the time with rated current;
 - With bypass;
 - from 0 to 55 °C without current derating
- 3) For temperature over the rated value;
- 4) With reduction factor;
- 5) For altitude above 1000 m (3281 ft);
- 6) For altitude above 2000 m (6562 ft);
- 7) Image merely illustrative;
- 8) For further information consult the product manuals SSW900.

